

ActInf Livestream #023.0 ~ "Embodied skillful performance: where the acti

Livestream | 1:25:29

all right we are live

welcome everybody thank you for joining

us live or in replay

this is actin flab live stream number

23.0

it's may 27th 2021 and we're going to be

talking about the paper

embodied skillful performance and i'm

daniel

i'm here with blue and dean so

thanks for joining you too

welcome everybody to the actonflab we're

a participatory online lab

that is communicating learning and

practicing

applied active inference you can find us

at the links that we have laid out here

this is recorded in an archived live

stream so please provide us with

feedback so that we can improve our work

all backgrounds and perspectives are

welcome to contribute here

and as far as video etiquette for live

streams we'll be doing our best

we're heading into june so soon in just

a few weeks

and we'll be having discussion number

23.1

and dot 2 on this paper so today it's

going to be about

background and context and then we're

looking forward on june 8th

and 15th 2021 to be talking about this

in a group setting so as stated

the goal of this 23.0 is to set the

context

for the upcoming 1 and 2 videos

we're going to be discussing the paper

embodied skillful performance

where the action is which is a

collaboration of hippolyto baltierri

fristen and ramstead

and just like the other dot zeros it's

an introduction and a first pass to some

of these ideas it's not

a review or a final word

we're going to go over the aims and

claims

the abstract roadmap and then kind of

walk through the figures plus a few of

these ideas because there are a lot of

cool ideas

and it's an embodied paper it's about

our movement and our behavior

in the world so hopefully that will lead

all of us to have some

fun ideas about what it means and where

it applies

and of course we're looking forward to

the dot one and dot two discussions

so you can write any questions during

this stream in the live chat or you can

write a comment

or find a way to get that question to us

otherwise and get in touch if you want

to participate

so to slide six

we're checking out this paper embodied

skillful performance

where the action is and before we get to

the aims and claims

maybe each of you any just top of the

head

thoughts what made you excited or
curious about this paper
i don't want to talk too much about this
because it was a great paper but
i again could go off into the wilderness
very
rapidly so i want to just kind of stick
to what the
what the basics of the paper are and
we'll worry about other ideas in
point one and point two for me sounds
great
thanks for that contribution blue i've
been just thinking a lot about um
you know related to this paper and other
work i'm involved in right now like how
beliefs and desires and intentions map
to action
so it was really kind of cool to see
this perspective
nice um so maybe blue
want to start with a aim of the paper
or i'll read it or sorry i was needed
this is the aim of this paper is to
discuss critically the limitations of
instructionist control theoretic models
of
skill for skillful performance so
that's the overall aim yep nice that
they said it
so clearly so that's what they're trying
to do they're bringing up
what's the instructionist control theory
model of skillful performance which
we're gonna
cover the background on and then they're
going to
show that that instructionist assumption

in its strong and weak form
is imbued in the optimal motor control
theory or omct
framework and then coming in to save the
day
active inference is going to be proposed
as an alternative
framework for understanding skillful
performance
and they then conclude by suggesting
that if we understand these generative
models
within active inference as formal tools
then we may be able to have an account
of skilled performance
that's based upon interactions rather
than instructions
so that's kind of cool because
instructions
are how computers work and education is
also
sometimes framed as being about
instruction rather than interaction
so here is the abstract and then either
of you are welcome to add a thought or
it stands alone when someone masters a
skill their performance looks to us
like second nature it looks as if their
actions are smoothly performed without
explicit
knowledge driven online monitoring of
their performance
contemporary computational models in
motor control theory however
are instructionist that is they cast
skillful performance as a knowledge
driven process
optimal motor control theory omct

as representative par excellence of such approaches casts skillful performance as an instruction instantiated in the brain that needs to be executed a motor command this paper aims to show the limitations of such instructionist approaches to skillful performance we specifically address the question of whether the assumption of control theoretic models is warranted then they lay out the three sections or four sections of the paper the first section of this paper examines the instructionist assumption according to which skillful performance consists of the execution of theoretical instructions harnessed in motor representations the second and third sections characterize the implementation of motor representations as motor commands with a special focus on the formulations from omct the final sections of this paper examine predictive coding and active inference behavioral modeling frameworks that descend but are distinct from omct and argued that the instructionist control theoretic assumptions are ill-motivated in light of new developments in active inference any thoughts on that yep dean so i think i think that that that there is a place for command and

control but it's not the
only place or the sort of all-in
all embracing place and i think that's
what the paper
sets out to try to show yes they don't
say
um there is no such thing as
um instruction they just suggest
that for talking about motor control and
for skilled performance especially maybe
there's a better way to frame it the
roadmap
is shown here so it starts with an
introduction
section they then introduce their um
whipping horse the scapegoat the
instructionist model
of skillful performance and discuss how
motor representations are playing the
role essentially of motor commands
in a sort of brain instructing the body
model
of motor behavior they talk about how
motor commands
are representational and we'll talk a
little bit about representations
as well and then connect the ideas
of motor commands and instructionism to
optimal control theory so basically
saying in order to achieve that kind of
optimal or even adequate control
instructions are being sent to the motor
department
of the university from a non-motor
department
like the nervous system in section 6
less control more action they try to
move the discussion from simply optimal

control

to a broader consideration of the

dynamics of biological systems

framed generally as in the case of

predictive coding and active inference

and then in section seven they discuss

how motor control

um again building up this argument that

it's about interactions

not just instructions they frame motor

control as being

a mode of sensory motor engagement with

the world

and even that word sensory motor kind of

suggests that there's going to be a

non-duality between

sense and action which brings us of

course to active inference

and the conclusion and also then after

the conclusion there's a box

where they define active inference and

free energy and some other

terms so that's the um roadmap

any thoughts on the roadmap or we'll

just keep on going

nice okay the keywords were laid out

by the authors skillful performance it's

right there in the title

so not too surprising that it's a

keyword they also talk about

optimal control theory instructionism

motor representation action-oriented

representations

and our favorite active inference so

we're going to

now meander through these keywords

and define it try to use the author's

own words to see how they

framed the keyword and then also just because we're having fun here in this trio we took it in a few different ways and brought up some related ideas and so i think this will be a fun walk through so i'll read the author's definition of skillful performance and then both of you please um so skillful performance in the paper is defined as as opposed to bare movements such as breathing and blinking skillful performances are intelligent bodily activities which harness knowledge about how to perform certain movements expertly and just to describe some of the expert movements that we see on this slide we have a basketball a sports ball being played with a free throw we have a pilot this is kind of an extended skillful performance involving a flight deck in a in the cockpit we have various styles of dance and performance and martial arts and interaction whether um touching somebody or not touching somebody these are all skillful activities that people can train their entire life to enable um the performance of so either of you i'm sure you can jump in on this um key term so blue first yeah i'll jump in you know i thought like of course these are all very physical

and like sensory motor but there's also

i mean i guess

all things are but like music i think

was brought up in the paper that's not
here

um and i think about stuff i've done

like you know and think about like

surgery right like like what a surgeon

does like they go that is a skillful

performance

it might not be in front of others but

like you have to just kind of know and

like guided by your kinesthetic memory

like any kind of aseptic technique lab

work i'm sure you've done some wet lab

work daniel like where it's like you

come in and it's like you just have to

flow through it yeah pipetting yep

they have robots for that now and that's

the joke how do you know someone's a

biologist

they can take the cap of the toothpaste

off and put it back on with one hand

that's what we do with bottles of

reagents so that

and it takes so much time like

to think about it like at first and then

you just do it without thinking about it

like it's it's

so there's this ironic like where you're

consciously like

trying to do it with one hand and not

touch the other thing and then

like after you do it for so many years

it just becomes second nature

dean yeah i just it was interesting to

me because i don't know when

when the the lines crossed and you go

from an autonomic
habituated response the part about
blinking
the ones that i don't think about versus
the ones when my
my spouse she looks at me and i wasn't
aware of it in the past but she was
using those blinks
in a certain way and was very effective
i i
went right over my head but she was
sending a signal and it worked
so is it i again it can come down
even to it can come to thinking and not
thinking about it and it can also come
to
sometimes what we notice and don't
notice
i agree and not to um generalize
too rapidly but perhaps active inference
can describe
the entire continuum of performance and
so we're almost dismantling this idea of
skilled performance because that
makes it sound like there's going to be
a distinction between skilled and
unskilled
whereas walking is a skill and there's
so many different
capacities and constraints and ways that
the body adapts to
itself and its needs so it's something
that you can learn and
anyone who's had a physical injury or
needs to go to physical therapy
it's relearning as well so i think it's
a big
way to talk about motor behavior but

it's almost like saying
if we can clear this high bar if we can
have an active inference account
that is playing piano or doing martial
arts then
surely we can have one that is more at a
unskilled level but it's still a skill
okay nice so then i'll play the video i
don't think you'll
see it but i'm playing the video in the
background of ilon
a plant hybrid hybrid uh plant robot
hybrid and this is from the mit
media lab so blue what made you want to
put this video here while it's playing
uh so i just kept thinking like they
kept using the the terminology motor
plant
and having seen this video like i was
like oh
motor plant that reminds me of this
motor
or plant that's attached to a motor uh
um
and the you know mechanisms that are
used for uh
motion in in the plant um which is is
awesome
so i just i don't know wanted to put it
in there and just see if
maybe our active inference active in for
plants applies
across you know all domains of life
yep very nice so it's kind of like we
know that active inference is a
scale-free
framework but also it sort of doesn't
have a

human-centered frame either so this is a plant that in the video you can see it's using its signaling processes to react to stimuli and move back and forth between these two lights so it's having it's kind of like a a person in a um mobility device you're giving it a new affordance by modifying its niche and so just like we're rethinking skilled versus unskilled we're also maybe rethinking who gets to have behavior and that made me think of a paper from 1989 by my phd advisor deborah gordon and a collaborator silvertown where they laid out a framework for plant behavior and they just plainly wrote out here we use the term behavior to mean what a plant or animal does in the course of an individual's lifetime in response to some event or change in its environment so it's like it's what things do you know just do it that's behavior and that's kind of how action is in active inference whether your active state is you can only move one finger or you can move all your fingers those are your active state affordances so we want to have a framework that is able to accommodate differences in individuals or even species in how they behave one thing daniel i think that the

the we the adding of the wheels and
you've hear you've heard that you you
well cliché the game it's a game changer
but essentially the rules changed in
that video

and that's and that's when the rules
change the behaviors can modify
and i think that's one of the things
that comes out in this paper later on if
we want to include that
we want to sort of inflate it up to that
level

nice and the skillful performance of
music you know if somebody is really
good at like

drumming on their own body or using
their voice box or beat boxing like
those are all sort of

on-board tools and affordances that we
have

people can clap really well and do all
sorts of stuff like that but

once you start talking about physical
instrumentation

whether it's a spoon or a violin or drum
set you're talking about

kind of modifying your niche and being
able to undertake skill performance
with your new tools so

again extended embedded and cultured
cognition the four

e's active inference it's right there

and that's why we want to think broadly
about skilled performance here and you
know invite

into this conversation whether a
technical

minded person or somebody who considers

some of their own behaviors to be
skilled performances how does this
remark on them
it brings us to what the authors call
the paradox
of skillful performance so they write
here's the paradox a seemingly
paradoxical relationship between action
and knowledge
action both requires knowledge and is
inhibited by knowledge
skillful performance involves both an
exquisite sensitivity
to cultural norms and situational
context but also
kind of requires us to put that aside at
least consciously
while it's happening how then to make
sense of the relation
between knowledge and skillful
performance
surely knowledge is involved in skillful
performance but
is it entirely in the form of a theory
that ought to be executed as a top-down
instruction
anything to add on this paradox
um i think that you can have
thinking is doing and then you can have
doing is doing and i think this is
talking to some of the ideas around the
quieting of the mind piece so instead of
always having
these commands coming through you don't
have to think about it so much anymore
so yeah i like the
idea that um you know knowledge is
it inhibits so it requires action

requires knowledge but it's also
inhibited by it like
if you're it like lends itself to the
kind of overthinking
process right like you're overthinking
it you're never going to get it it's
like um
i think someone was just speaking on
tuesday about like holding the butterfly
was it stephen
you know like it's like when you grip an
idea like i want to
ride a bicycle or play the piano you
have to hold it
and like let it rest on your hand but
don't like squeeze it or you'll you'll
kill the butterfly
and that kind of like reminds me of this
knowledge like if you're overthinking
the process of playing the piano like
you're going to get it all wrong
yep if you grab the handlebar too tight
or grab your dance partner too tight
and then when you take it into the area
like post poetry
recitation or playing a classical music
piece
are they thinking about the sheet music
is their internal
representation or knowledge what is
enabling that performance
that's kind of what we're going to be
exploring so they motivate
that paradox as being something that has
to be explained why do people choke
when they think too much about their
golf swing or their tennis swing
and we'll come back to flow and thinking

about that later on
there has to be like this correct
relational distance
from the task that you're trying to to
perform
and i think that that is you know
embodied by this
sentence and action requires requires
knowledge and is inhibited by it
you have to be the right distance away
from it like if you're too close it
doesn't make sense and if you're too far
away it doesn't make sense
yep cool yeah i don't want to add one
other thing like yesterday i was walking
in the park and there's three kids that
were zooming down the hill on their
scooters
and the first kid was making the sound
we
like this like not we as in she was
having a good time but
she felt she had to put the reflective
and the analytical on
it because she was going first she
wasn't she wasn't thinking about how
much fun she was having
she was thinking about oh i'm supposed
to say this is exciting even though i'm
really not sure if i'm gonna go off into
the trees or whatever
and so it's it's fascinating how we kind
of get habituated to this idea that
in a cultural setting we have to let
others know
we're having fun when really what we're
trying to do is not
fall off the scooter right like that's

that's a big part of this this
overthinking piece
nice and a roller coaster where you
can't change
the way that the roller coaster is
working but you know it just erupts
people
screaming or something like that what is
going on there
optimal or laughing in a stand-up comedy
situation
so optimal control theory
so that's oct and they define oct
is a field in mathematical optimization
that deals with finding a control
for a dynamical system but they spend
most of the discussion of course with a
focus on motor behavior
and so they introduce the ocmt
optimal control motor theory and right
omct is a label for the modeling tools
used to study
motor behavior and its neural processing
we target
specifically omct to show that it rests
on the instructionist assumption which
we're going to cover
that the brain literally contains and
leverages explicit instructions for
movements
so what did either of you think
about ocmt
i just thought it was really interesting
because i would imagine that there would
be some emotion on the part of the
spotter not the person on the
handstand if suddenly his cigarette fell
out of his mouth i assume that's a

cigarette that's in his mouth
but he wouldn't even be aware of the
fact that he lost
something that he was losing optimal
control
because his focus was elsewhere in front
of him
so again you you have to kind of you
have to kind of take this idea of skill
and make sure that you understand
what you're talking about because uh
yeah
there's probably a lot of skill
perceived in the person doing the
handstand
but there's also other skills going off
there that we don't give as much
attention over to
yep there's the skill of standing up
which is a dynamic
process there's holding an object in the
lips even of the observer
and then there's the spotter's skill to
know
when to intervene and when not to
intervene and
half the battle is knowing when not to
move
and when to just let it ride and trust
in the partner
so it's kind of cool that when people
look at this photo
they might immediately be highlighting
this sort of
difficult looking task the stacked
chairs and whatnot but
it's an idea that would help us explain
this whole situation

as active inference agents engaging in
motor behavior potentially
well it just asks us to redefine what
command and control is
basically nice
so how about instructionism so on the
left is a quote which
we're not going to read in full but
they do right down there at the bottom
and um i know blue that we'll have a
little bit to say on this from the
genetics angle
but they write following wheeler and
clark's 1999
paper yes that andy clark we will refer
to this position as instructionism
which is the idea that skill is
essentially of a theoretical nature
mediated by knowledge we cast
instructionism
in terms of explicit instructions that
is forms of knowledge that directly
guide performance
like okay i'm supposed to run over to
the left side of the stage when i hear
the timpani
boom applied to skillful performance
the instructionist assumption prescribes
that instructions are harnessed
inseparable
structures such as beliefs and then on
the left side
is the abstract and title of wheeler and
clark 99
which is actually a paper called genic
representation
about genes reconciling content and
causal complexity so blue any

thoughts on where genes come into play
i mean it's all right like uh
there's penetrance and there's you know
all of these things
that come into this you know genetic um
relationship and so i think it's it's
interesting to think about and also
i think as the authors point out that
there's you know the genes
aren't really the instructions for
creating protein
right like they are just the fundamental
blueprint
so they don't have such a direct
relationship with with action
they're just like the belief system
they're not the intention
i don't know we're gonna spiral back to
the mapping between
genes and motor representations in a
little bit
but just cool that they drew from a
different area
of philosophy and science to be building
their instructionist point but
for the next few slides let's focus back
in on the motor behavior
so here we highlight a few different
attributes
of instructionism so if somebody is
asking well like what is instructionism
or what are its characteristics or how
would i know if
a theory is being cast in an
instructionist frame
here in red we have um a few words like
separability and its cousin modularity
we have representational thinking which

we're going to come back to as well
and normativity meaning how things
should be done and these are quotes
saying
instructionism saying that instructions
come in the form of separable structures
such as beliefs
so that implies separability modularity
and then they also
say that instructionism suggests that
performance is enabled by
motor representations which harness
knowledge about how a specific skillful
performance
should be executed like if the
instructions were to
have you fall over that's not how you
get to a walking person
so it's about how things should be and
then if there's like something on your
leg
grabbing you down it's about how you
should escape so
here's kind of a um an example
of instructionism with a person chopping
herbs
and it's almost like you could imagine
step one you know first instruction
grip the herbs with one hand step two
grip the knife and then step three is
gonna be like move your arm up and then
move your arm down
move it up go to the side drop it down
again so that's sort of
a simplification but it's highlighting
how it's representing
it's saying what should be done and it's
giving separable

instructions like a recipe and so it's kind of a knowledge knowledge-driven pseudo-code or recipe for getting a motor behavior done okay first you're gonna just do this then you're gonna do that um and they basically again highlight and cite suggesting that most accounts of motor representation have in common that they're representations of knowledge that get translated in a way into motor action so that's the instructionism any thoughts on the knife slide so you know having watched a chef do this like when i used to work in restaurants and then like trying to teach my children like my daughter's nine right like it is horrifying i'm like you are gonna cut your finger off but like you watch a chef and they can do it they can be talking to you and you know i mean like doing 10 other things and chop chop chop like not even looking or paying attention just driving by by sheer kinesthetic memory uh and then my daughter's like girl it's really scary yep but if um the instructions were too strong then the individual would never learn how to chop without the direct oversight so it's sort of uh yeah one thing does this not demonstrate clearly

that knowledge and information are not
the same thing
what do you mean by that well i mean
both are portable
but do we do we assume that they're the
same thing when in fact
what knowledge is is it demarcates
how much mental time travel we can we
can span
as opposed to information which can fill
up volumes right
but it's not the same thing one is that
one is literally that we open the lid on
the brain
and we pour this stuff in but when we
when we
unpack that it doesn't come out the way
that it
it went in it comes out and the
knowledge comes out as
something that we can relate in terms of
time spans that's why we've had a lot of
these conversations in a lot of these
a lot of these livestreams that i
participated in there a lot of people
now are coming back to so what role does
the hippocampus play
is it a spatial thing or can we
segregate that out from
time or are these time cells
as opposed to space cells i don't
i i'm not going to go much further on
this but
i believe that this slide shows once and
for all
that we should stop thinking of
information as knowledge
so i've thought a lot about this dean

about um
the relationship right between like
information
data knowledge and
wisdom and like i've thought about this
in kind of like big groups and come with
to like touch on some tentative
definitions like data is information
that's useful right in some way
and then knowledge is you know the the
the application of of the data or or the
the interpretation of the data
and then like where wisdom is kind of
knowing how to
apply the knowledge right so it's kind
of like a a succession
of but from you know information to
wisdom i i look at it in like a
like a hierarchical model do you see how
people that
are having difficulty maybe they're
they're
addressing issues with dementia or
alzheimer's and when they're
when their ability to mentally time
travel collapses
so does their knowledge
very interesting at least
from a working memory and fluid ability
to
integrate and interpret in those things
now again
i don't want to go off into the weeds so
i'm not going to do that
but i think one of the biggest problems
that
this paper is pointing out is if you
believe that

information and knowledge are the same
thing you might want to go
to a place that says command and control
is what's actually happening
my suggestion is this this blows that up
right here right now now maybe i'm wrong
but i don't think i am so interestingly
dean i don't know if you
know this or not but alzheimer's and
dementia patients
um like while they won't even remember
the names of their children
they do remember the lyrics to songs
right and and like they remember music
in a way that is very different so
perhaps when you get to
the level of you know skilled
performance and and i know like the
active inference is going to
kind of squish all motor in in a minute
we're going to squish it all together
but when you think about skill
performance perhaps
that knowledge is integrated in a deeper
way right like you might be able to give
to get an alzheimer's patient if they
were physically capable um
to ride a bicycle like they still retain
that knowledge
without having to do some kind of
command so there's maybe like
some chain in the link is broken there
it's interesting to think about
yeah i mean i've got i got i've got
first-hand
experience because i've got family that
are that are addressing this
and yeah so i know exactly what you're

saying and again

i don't i don't want to i don't want to
carry this off too far but i think it's
i think it's worth contemplating what
knowledge is

relative to what information is that's
all

awesome points and for moving bodies
bodies in motion space and time
are going to be deeply intertwined
you're always somewhere somewhen
but we have the clock and we have the
gps and so it makes it seem like you
could separate those out but not for the
body

not for the embodied agent

so there's a few uh types of
instructionism

that they then flesh out so that was the
big category it's about instructions
and they distinguish two kinds of
instructionism the strong form
and the weak form the strong form
is the claim that neural representations
so specific like firing patterns and
neurons and

things like that completely specify on
their own

the specific movements to be executed so
if like you were recording from the
brain you could just template out
a specific firing sequence or a specific
pattern

and then that would be like the contract
bicep now signal

and the weaker form of instructionism is
a more modest claim

that among the many dynamically coupled

systems that generate skillful performance
so the neurons in the bicep and everything in between and different regions in the brain
so all those dynamically coupled systems there is a special kind maybe you didn't capture it maybe you don't know what it is but there there
ought to be a special kind and those are the structures internal to an agent that are responsible for encoding information or instructions that are relevant for action
given the background of everything else that is enabling that role
so in this modest account motor representations play the role in the generation of behavior
analogous to genes in the generation of phenotypic traits
so the strong form is like um would be like as strong as saying that the protein was hiding inside of the gene and then a dna gene and then it just popped out
whereas the weaker form is saying there's a lot of dynamically coupled systems
and um this weak instruction is something that's distributed maybe even among them but that's the part of the dynamics of the system that are relevant for action so strong and weak
and it turns out that they're going to um

have something to say about both them so
let's go
to is that good anything to add on that
one
yeah so here um
they're going to highlight the weak
instructionist claim
but as you can see from the bottom of
the slide they're going to argue that
neither
kind of instructionism is warranted so
whether you take a strong or a weak
approach
they say that you're still making the
same type of error
although maybe you're making it more
egregiously or not
the weak instructionist framework for
motor representation
says that skillful performance is the
result of an orchestrated process
spanning components in the brain body
and world but some of these components
especially those in the brain play
specific
explanatory irreducible roles of
encoding explicit instructions for motor
performance
so that is the instructionist mindset
and
for those of you you know inside or
outside of the active inference
game is that your implicit
understanding of like what is happening
would you say that there is
something like an instruction for pick
up the finger or
is there something like i should be

peddling in the brain
because that's the exact kind of notion
that they're going to be seeking to
challenge
so that's what's on the table is any
idea no matter how strong or weak
that instructions can be something that
is
distributed throughout the system are
useful for skillful performance
that's what they're going to be
critiquing blue
um so i just really wanted to kind of
loop back to the gene
um versus protein right so so
a dna gene gets transcribed or
yeah transcribed into mrna and then
translated into protein and so
but you can still like have
environmental modifications on that dna
gene
which i think that's kind of a neat um
like
way to kind of think about it like you
can have methylation
um epigenetic modifications or like you
can also have
like suppression of the gene like by
rolling it up in the chromatin so
so many different like environmental
factors can still map on to that dna
gene to modify
it so that the instructions are somehow
different
cool so let's go from
instructionism of any kind to
representation
so if there is some sort of instruction

in any part of the system whether it's
in one neuron or
a set of neurons or it's between neurons
and glia and all kinds of cells
if there is some kind of representation
what is that representation
so here's a quote from the paper with a
red box they wrote
what does it mean for a thing to
explicitly represent some state of
affairs
it is common in the philosophy of mind
to argue that representations
involve modes of presentation
it's right there re-presentation
and that um sort of made me curious
about following to this
phrase uh not sure if that's the
pronunciation 1892
citation sense and reference
and um there's a few pieces that
were kind of interesting about this 1892
paper
so one thing is that it's written
the question of truth would cause us to
abandon aesthetic delight for an
attitude of scientific investigation
so that's maybe even like the spirit of
what we do
here we try to be clear when we're
talking about what we want to be clear
about
but we always want to leave the door in
the window and the roof
open for aesthetics and for art and for
that kind of delight hence it's a matter
of indifference to us whether the name
odysseus for instance has a referent

so long as we accept the poem as a work
of art it's the striving for truth that
derives us
always to advance from the sense to the
referent
and that is what um is known as the
semiotic triad which is the relationship
between
the symbol which is like maybe dean
actually what would you say about the
semiotic triad yeah
it's it's back to piercy and stuff
charles sanders pierce what i think is
really interesting
again is that you can have a content
policy
so you can have information in
representative form
but knowledge is actually a change
policy
it's the advance from
sense to referent so you don't want to
mix the two things
up one is content the other is change
and if you want to say that apples are
oranges you can have that
i'm not saying that our minds aren't
capable of doing that
but you can't impose it and that's what
i think that
maybe i look like a guy from the 1890s
because i just have a lot of respect for
everything that they talked about 100
years ago but yeah
i mean it's right there you don't want
to fly
you don't want to crash your your your
airplane into that mountain

is my point mount sense um and then also
that
um note 6 right after art is it'd be
desirable to have a special term for
signs having only sense
hmm what would it mean to have sense
like it compiles in a sensical way but
it doesn't have
reference if we name them say
representations the words of the actors
on stage would be representations
indeed the actor himself would be
representation well
what if all the world's a stage and then
who's not
representing so these are some really
cool ideas about what
is being um communicated symbolically
when we use symbols like language so
representation is a big topic and we
talked about
the representation wars and about
representation and active
inference beyond internalism and
externalism it's a topic we're going to
probably be coming back to forever
because it's such a rich
area let's go back to motor
representations though
so that's what a representation was sort
of broadly
but bringing it back to the body to the
motor behavior
they cite a 2017 paper that provides a
definition of motor representations
this account is basically coming from a
combination
of pragmatism practical representations

as well as computational neuroscience research and they highlight three key attributes in this modern research on motor representation the first is that those motor representations um represent objects and situation in terms of their properties relevant for action so like a great example is like a cup it has a handle and like your brain will do a different thing when somebody grabs it from the top or when somebody grabs it um with a weird way of grabbing it so we see objects for actions we see things to sit on and things to grab we don't just see the pixels or anything like that a second key aspect of motor representation is that they're informed by or contain implicitly some knowledge about the body's biomechanical and kinematic constraints so you'll see something for gripping that's like grippable you won't see like a 10-foot pipe as being something you could grip but then as it shrinks given the implicit or explicit knowledge of what the body can do it becomes something that's grippable and then third they say that motor representations at least usually serve the execution of transient

movements

so in that knife cutting example there's
a motor representation for like
contract up and bring start to bring the
arm up

but it doesn't need to be the
representation of like the entire
story arc it just has to be a transient
executable

um motor action so that's
again motor representation it's within
this instructionism

frame and then dean you had a really
interesting comment down here
well all i wanted to point out was that
um

so the active inference allows us to
well what they're going to talk about
what we're going to get to is the the
the gender that generative model
can be inverted but essentially there's
top down on the and there's bottom up
and i think that's what needs to be
included but it's

i in the last sentence of that little
comment world to mind and mind to world
that ability to sort of go

bottom up and top down assumes that a
capacity to differentiate what actually
what actually contains um

and then what what is contained in a
practical sense

beyond a thought experiment we can all
do thought experiments

we can all reflect and analyze prior to
and and post

but when you're in that otherwise and
you're actually carrying it out

you're sort of going beyond that thought
experiment so

how do we how do we address that when
we're doing our motor representing
is what i basically wanted and that's
just from the paper yeah

basically vc's decomposition cool
we're having so much fun but we are
going to i think try to get through the
rest of the slides

here's one way that motor
representations are used just in the
literature

they use fmri they find a pattern that
correlates with a certain motor behavior
and that is discussed as being the motor
representation

so that's how scientists are using the
term

we can think about motor representations
as being motor commands

so i think blue did you um make this one
and want to add something here

yeah sure so um this motor command i
thought was interesting because

it's like you have to have the forward
and the reverse

in this in this motor command so you
have to already know

the outcome that you like you have to
envision the outcome in that said

environment um and then work backwards
from there so i'll just read this part

in blue

it says that this entails an inverse
inference problem

which requires working back from the
desired sensory consequences

eg desired visual and proprioceptive
sensory feedback
to a specification of their motor cause
in an intrinsic frame of reference
i.e a set of muscle activations that can
generate
such desired consequences so i just
thought that that was an interesting
um way to think about it and i always
loved the picture of like the
this is like the motor representation
like the literal
like motor representation on um the
cerebellum and so it just
shows or or maybe this is cortex um
so this is literally like how it maps to
the sensory cortex
like your which part of your piece of
your brain literally maps to each piece
of your body
it's interesting and it reminds me of
coaching
cues like lift lift lift or push
something like that you say what you
want to see
happen lift the bar or something like
that you don't
have to say every single cue so that's
sort of a semantic way of engaging
with what's happening across the brain
and
across the body and to bring
the motor to action before we go to
active inference
they um talk about action-oriented
representations as being
the motor representations that are for
action

so anything on this blue
yeah but it's another example like
breathe in pretend you're smelling a
flower
which one conveys more of a story and
that reminds me actually of dave
snowden's
address not so long ago where instead of
just asking someone you know tell me a
story
about san francisco you'd say
a friend only has two minutes and they
want to know the single thing that's
most exciting to visit when they go
that's a very rich setup for somebody to
tell a story in
and so it relates to giving suggestive
cues
that are richer than just contract your
bicep but you could give a cue
you know bend it like bechham or
something like that you can make a
metaphor or you can allude to something
that actually
invokes more activation you know
would be interesting can people deadlift
more if you say all right
lift up the car you know somebody's
underneath it
go for it or you know you're going to
win this competition
if you do this those are like very
action oriented motivators dean
yeah actually i put this in here this
slide because
one of the i did a lot of coaching in my
life probably 30 years of it
and i think one of the things that i

found as an as an effective coach
is you want to think of it as as
any puzzle piece is missing back to that
back to that's i don't want to go back
up to the slider the guy was balancing
on the edge of the building but again as
a coach
there'll be things that the person who's
performing that motor skill
will be thinking about and what what
what an effective coach tends to do
is try to bring into
into the the realm of what they are
focusing on
without interfering with that without
adding a bunch of cognitive
issues that again get in the way of
of of doing making it an overthinking
thing if you can breathe if you can
remind a person to breathe out
or exhale just before they do something
that they would do that autonomically
but if you can actually get them to
think about that that again
that's a mind-calming exercise that's
actually a pretty effective coaching
tool
cool and i think we're going to come
back to that with instruction versus
interaction
exactly then we turn to active inference
yay and they have box one
in their uh paper at the very end where
they define active inference
was it a reviewer did they want to
clarify we'll find out soon
and they basically write that a system
can be described to engage in active

inference

in the sense of performing belief

updating and acting such as to fulfill

prior preferences about observations

because

in active inference there's kind of two

things that the system can do

it can act or it can infer it can update

its action states

it can update its internal generative

model of the world those are sort of the

two

levers that it has to play with because

it can't directly control the

observations

but those two levers action and

inference are being

played with in service of reducing

uncertainty about observations

so that's some of the key attributes of

active inference

and they suggest that active inference

provides a mechanism

to derive the dynamics of sensory and

active states which are these two

blanket states on this image such that

they minimize

a variational free energy functional so

we're not going to go into that

now but there's basically an equation a

formalism

that trades off model complexity how

many parameters it has

with model accuracy so real systems are

sort of

on this razor's edge where they want to

have well-fitting but simple models

and then the tuning between you know

great tasting and less filling
the tuning between fewer parameters and
better fit
that is where active inference agents
are surfing
anything on active inference or let's
just see how it gets
used
so the paper draws extensively on first
in 2011
the figures and a lot of the citations
so it's worthwhile to
actually go to that first in 2011 paper
if you're curious about a deeper level
and here from the um
skillful performance paper they
basically
oh this is actually sorry this is from
uh first in 2011.
despite the compelling simplicity of the
optimal control approach
he's saying that it's wrong for two
reasons
the first is that basing
on the ideas from information flow
he's going to suggest that motion cannot
be specified by a single
value function there might be many
different ways to achieve a goal
so it might be hard to rank or even
compare
totally different ways of achieving that
goal on
a value only framework and then secondly
optimal control theory assumes that
movement is caused
or determined by value like motors
motor actions are valued differentially

and then
ones that are more valuable are
undertaken however
value is an attribute of the states that
are caused by movement it's a
consequence
not a cause this means that the real
problem is to understand the acquisition
and realization of beliefs that cause
movement
in other words to understand motor
control in terms of inference and belief
and then another really interesting
piece was to describe optimal control
even way back when in 2011 as
active inference with three
simplifications so the first
simplification
is saying you gotta combine action
inference together
you could simplify and consider only
action or consider only perception
but optimal control is not um as
is not dealing with those two in a way
as integrated as active inference
the second simplification has to do with
the delays
in the signals so that's a little bit of
a motor specific
a more neural perspective on motor
control
and then the third piece which is
perhaps the most salient
is to realize that value alone and cost
functions
doesn't get you all the way to embodied
action
but embodied action is more consistent

with thinking about how
organisms have prior beliefs about what
kind of motor behaviors are likely for
them
and that is a broader framework than
just value
so that's first in 2011 but there's of
course a whole paper there so if you're
curious you should
check it out okay figure one
want to take a first pass on this one
sure i'll read it so they
just like um we had talked about earlier
you have the extrinsic frame of
reference like what needs to be done
in in this environment right so the
motor plant is the physical
motor system like a limb that executes
the task to be
performed and that's the
plant kinetics equation here
and then physical movements of the motor
plant in turn
generate sensory information this
information is conveyed
to a state estimator via a sensory
mapping so that's the sensory mapping
process the next equation
and then there's the the state estimator
in the middle
and over on the right the forward model
it says the function of the forward
model is to improve
the execution of action by helping
to finesse the inferences of the state
estimator and you can see that there's a
feedback with
the state estimator both forward and

backwards

uh and then the reference copy says the

optimal controller oh yeah that's this

next part down here at the bottom

um the optimal controller maps desired

trajectories

specified in extrinsic coordinates to

muscle movements

ie to changes in muscular states

specified

in terms of intrinsic coordinates um and

then that

goes back loops back to the plant

kinetics

yep so this is the whole um forward

model in motor control so this is

basically

um the optimal motor control theory

framing of what is happening in the

brain

um starts with a control implication

like control theory or cybernetics

motor behavior occurs in the the plant

the motor plants the

the infra plant but not here and then

that gets mapped through sense

to a state estimator where is my hand

and then that is going to get in

feedback with a forward model

that then says oh your hand is too

gripped

it needs to relax and then that feeds

back to the optimal control

equation so whether these actually are

instantiated

in modules of the brain that's like the

super strong

take the weaker take would be or the

instrumental take would be that we could
model as if it were this way
okay now they wrote about
recalling this uh now they're gonna
bring together the
motor representations and this
optimal motor control theory way of
forward and reverse modeling so this is
an intrinsic to extrinsic mapping
intrinsic is the motor commands the
representations internal to the system
and the forward model goes from motor
command to bodily motion
so that's like the instruction is like
relax
the hand okay and then it gets
implemented but interestingly just
thinking
relax your hand semantically or
syntactically it doesn't open your hand
right or you know play beethoven doesn't
work
quite like that so the instructions are
not
just what we hear in our inner voice
but that is the forward model and then
the inverse model
would be the mapping from bodily motions
back to the commands
and to now connect it to their citation
of the wheeler and clark 99
it's like the dna genes it's a story for
another day
why the gene concept is a little bit
more nuanced than just that but again
that's another day the the genes the dna
genes are like
intrinsic and forward genetics

is the study of how changes in dna
influence
phenotype and reverse genetics is going
from an interesting phenotype
and then asking what were the genetic
changes that caused it
so it actually helped me as a geneticist
to understand what was happening here by
thinking about forward and reverse
genetics
and then thinking okay so then the kinds
of arguments
that people make about how genes go to
protein
are like the arguments that people go
from motor representations to motor
behavior
i.e that the information is in the genes
it's in our dna
i mean you see that on billboards on the
freeway it's in our company's dna
it's in our motor representation the
motor behavior
or is it so um they talk about forward
and reverse model or inverse models and
that's again like kind of like
forward and reverse genetics any
thoughts on that
i was just thinking about self-driving
cars and like where's that like motor
representation
and then dean you wrote a really nice
point in the uh
comments well i just i just i honestly
why should i feel bad for omct but i did
because it was at this point i was like
man and and all the people that are sort
of really invested in this they're like

they're vested in this
this is how it works and again it was
kind of like
okay if you want a set of commands like
i understand a bunch of people that
program and they've and they do that
sort of thing and so it's an easy
it's an easy transference to say that a
content policy
is also a change policy but how do you
write out a change policy
yep and i think this is just as you
highlighted it in the notes there it
says
um the omct is inverting the forward
model
so they're literally looking for an
inversion from the bodily back to the
motor
this is contrasted with active inference
which replaces the inverse forward model
pair
with a generative model that expands on
the forward model
to harness probabilistic beliefs about
the expected sensory causes of action
so rather than using a separate inverse
model to infer what is appropriate
action
you have basically just a generative
model that's integrating
that feedback loop you have
yes so um this is
uh an idea that i think we've heard
steven talk about too the sandwich model
the classical sandwich
it's actually called the classical
sandwich of cognition

so this is kind of a classical sandwich
for me that looked
tasty um one of the
every every stream with dean you've got
to have like a sandwich or a hamburger
[Laughter]
um a scentee burger yes
yeah this this is a a quasi sandwich
a major assumption is in omct
is the often implicit reliance on a
sequential and modular architecture
of perception cognition and action which
is the sense model plan
act or the ooda observed orient
decide act paradigm or the classical
sandwich
so it kind of like sees these epochs of
action where first a new stimuli comes
in that's observe
and then that gets integrated with
cognition
um orient and decide in the ooda loop or
in the cognitive framing about modeling
and planning
and then that gets translated into
seems fair that is the um sense
think act cycle and that's the classical
sandwich so i don't know what food
active inference
would be but it's not going to be a
classical sandwich
okay maybe we'll leave that as an open
thought question
if active inference were a food or a
style
of food what might it be um
okay and then um that takes us
so again they're just these are like

they're sort of like two
teams we have instructionism motor
representation
omct sense plan act classical sandwich
that's not active inference active
inference is gonna be
in contrast to all of them and hopefully
getting some of the limitations resolved
um do you have anything to say about
this um
okay in this section less control more
action they're gonna actually
move we spent most of this video
highlighting
what that integrated set of ideas was
the non-active inference ideas about how
all those topics were related
and they basically here are going to
highlight
how omct does have an instructionist
assumption and how that relies on
separable forward and inverse models
rather than an integrated generative
model and how that
relates to motor commands rather than
something else
so finally we get to active inference
and how motor control is going to occur
and um we'll
we'll juxtapose these two figures in a
second
but um this is again a similar drawing
in fact it might even make sense just to
show um figure 1 and figure 2 side by
side
because at first i looked at them i
thought it's the same brain
and i'm kind of seeing the same like

shapes on it
but we can go to 30 and look at the two
differences
so it's color coded left is the figure 1
omct figure 2 is on the right with the
active inference framing
so we can see that plant kinetics and
sensory mappings
in red are found on both sides we also
see that there's a forward model
on both sides and that there is an
optimal
control element in blue on both sides
but it's that state estimator
at the nexus between the sensory mapping
and the forward model and the optimal
control there's the rub
something is rotten in the state
estimate of omct
that's what hamlet would have said
because there's something about
estimating the state
like that's where my hand is
that isn't happening in active inference
and that's really this key and
interesting piece
is that you can even look at the way
that the models are connected
to see that there's kind of like this
fundamental loop between the generative
model
forward model in active inference and
sense in action
getting spun off of that loop act infer
act infer that's active inference and
here it's like
perceive estimate state
reverse engineer value value-driven

assessment of optimal control that's a
quite

different layout for cognition this is
the one that's consistent with a sense
think act loop because

in that think as a separable module
you get state estimators how else would
you translate

from the perception to the action well
in active inference

we kind of have an answer for that at
least we have a model

framework that helps us get at it
anything to add on the figure or we'll
continue to how it challenges okay great
so

how does active inference challenge omct
either of you want to say something on
this one

so since i made this slide i liked this
um comparison and these two like
sentences really kind of stuck out for
me

um so in the omct which is linear
quadratic gaussian control that's this
lqq model that i'm going to mention so
it says this means that the canonical
controls

generated by the lqq models cannot
reduce or even

increase a system's uncertainty in the
future

i.e actions can only be instrumental and
have no

epistemic effect on future state
estimates

so i thought that that's kind of really
nailing at

what's the fundamental thing that omct
doesn't do that active inference is
going to allow for
so it says with active inference the
core idea then is rather than
select an explicit motor command the
organism
infers what it must be doing under the
assumption that what it must
that what it does must minimize
prediction error so there is that room
to reduce uncertainty in the active
inference model i just thought that was
an important distinction
dean anything on that no i think i think
blue
thanks for throwing that in there
because i think one of the things
that we have to sort of address is
do do all actions follow commands
i think the active inference says no we
can act first
yep as opposed to always following some
sort of a prompt
and i think that's this this slide
really spells that out clearly
yeah and this piece right here is that
about how actions can only be
instrumental and have
no epistemic effect so you need a total
bolted-on second layer in a value-driven
and value-evaluated model of motor
behavior
you need a second layer to explain play
to explain
exploratory behavior curiosity and just
like sort of
not caring that much whereas in active

inference

those are natural attributes of our
fundamental model

okay i think i think the omct

naturally constrains us if you're if

you're in the business of learning

it would be great because then everybody

would have to follow

there would be no other option but to

follow

but i think what active inference does

is say it takes the lid off that

says no i can model

first there there's nothing to follow

i'm not stuck

that reminds me almost of like the

teacher is the person who could get a

100

on the test so i'm up here on the top of

the mountain come on

75 is is okay 76 is better

come on one by one let's just let's go

over the questions you got wrong

so that you can get them right right

where does that come into play visually

so

sorry for the uh slightly busy slide

earlier versions were even busier

um active inference framework

comprehensively challenges the optimal

control theoretic

approach to sensory motor behavior in

particular the idea

in traditional omct is that behavior can

only

basically you can evaluate the

performance of different behaviors with

a single number

a scalar that is defined and tracked by
a value function
so this is sort of a traditional hill
climber approach
like if you want to get to the top of
the mountain then
the higher up the mountain you are the
better it is the more value
of that state so here's an agent on
landscape
and it's going to kind of take a ruler
and then it's going to
look at where it is locally and then
it's going to follow uphill it's like a
greedy
value pursuing agent whereas an active
inference
it actually draws on thermodynamics and
information flows
as suggested earlier to take the total
current the total flow and separate it
into two components one of those
is value so you still do get exactly
a separable component from the total
flow that represents
that ruler going uphill
but there's also a
vector that is like resting on that
ruler but going at a flat it's an iso
contour
it's kind of like when you look at a
topo map for going hiking
and that is what
oh yep that's the um the physics of flow
embedded in active inference show that
motion in a biologically realistic state
irreducibly includes two orthogonal
kinds of motion so those

two rulers again one is going on
whatever the angle is of the hill
and the other one is going on the flat
part those are going to be
always like orthogonal to each other
like at a 90 degree cross
so there's an irrotational curl free
component
that's which movement will take me just
locally
up the hill and then there's a
solenoidal
or a divergence-free component that
circulates you
on the same elevation
the curl free component allows the flow
to climb a gradient towards more
valuable
or probable states the solenoidal
component
specifies a flow around an iso
probability contour where all entered
states have an equal value or
probability
so there is an iso contour for this
agent on the left
but it's not calculating or tracking it
in active inference
we could turn the knob so that it just
pursues value
but we also have a framework for that
sort of play
and who on the rugged landscape is going
to get to the top of the hill
an agent that's able to switch back and
forth
or you know shade between these
different ways of working

it's like you climb up and then you're
at the top of a local peak
and then can you circulate a little bit
um so that is part of how information
theory
and information flows play in active
inference let's take
information flow from a sort of
quantitative
ontology of information flow like first
in 2011
and in this figure to this version of a
flow state so i'm gonna play
this video on repeat blue while you
describe what you're excited about here
cool so you know just upon reading like
the title
of the paper um skill performance i was
like oh
active inference we're going to talk
about the flow state and i was like
we got to flows but i kind of was left
like wanting to talk about um the flow
state
and so this is like the preview from the
disney movie soul
uh where everybody has like their thing
and they can
you know jump into a flow state um and
it's just showing funny parts from the
movie
of people who for people who have seen
it or not um it's very cute
but but in that flow state which i think
we've all kind of
experienced it is like the skilled
performance
where daniel and i were discussing

earlier like you can just do things
without thinking about them
and so i was you know reminded of
the um paper that we did toward a
predictive global neuronal workspace
um by chris white and ryan smith and
that was stream
18.0.1.2 if anybody wants to check it
out
but they're there they mapped um they
built a model of the visual
consciousness right and so we talked a
little bit about consciousness
and specifically phenomenological
consciousness like
where you're able to report like i
saw a red bird or or whatever like you
can say i saw a red bird and we talked
about
in that um video how the model didn't
apply for
or it like for um inattentional
blindness right it couldn't it couldn't
account for
when you see something but you can't
report that you saw it right so you have
to have this
this phenomenological consequence or
consciousness and so i was thinking
about the flow state
and really like is there like a motor
consciousness like right like so
you know i'm conscious that my feet are
on the floor and like this is something
for people who meditate like a lot of it
is like awareness of the body and where
the body is in space and so
it's sometimes you can use

proprioception there but
am i conscious of where my body is where
i want to put my body or
can i just do it like is it like
inattentional blindness like when you're
in that flow state like when you're
dancing
you know like you don't you can't report
like oh i did a pirouette
and then you know or whatever like you
can you can't
verbally report what's happening with
your
motor consciousness in a a lot of times
especially when you're in that like flow
state so is that like
similar maybe to inattentional blindness
anyway it's just something that i was
thinking about
um and i think that there's an
interesting contrast between the way
that this paper
models the motor active inference versus
the visual active inference
that was modeled in that paper and it
almost reminds me of
like a sports you know or like a
trackmate and sometimes the reporter
okay how does that feel you just won the
game
the athlete just their body says it all
or the performance says it all and it's
like
and then and then what and it's like as
if they were running alongside trying to
get the play-by-play
from the person who's actually in the
action

they're not going to be able to report
it in the same way it breaks them out of
the flow state
to have to verbally represent it which
is why
in sports commentary sometimes you have
like a conversation happening
between like the more maybe pragmatic or
descriptive
representation and then a more like
color commentary
because that plays both roles of
giving the information about what is
happening but also some of that
excitement and connecting the dots but
the person you can't ask
in the moment is the one who's actually
doing the action
so that's cool yeah yeah often times
wondered if
after one of these performances the
reporter got the response well here's my
physiological state
i think that would be awesome right now
i'm just feeling like my right knee is
i can't move it anymore and then they
just
dropped the mic and walked away right
there's my tooth drop
nice maybe in the future they'll be
wearing you know physiological sensors
and that'll say it all right
so what does active inference sort of in
closing here
what is being added or contributed by
active inference
well it's a formal model of motor
control

it helps us understand online
no not like this live stream like real
time motor adaptation to
an environment and thinks about that
environment in terms of a
um not necessarily symmetric but
bi-directional relationship between
agent and environment
and brunberg and constant and many other
papers
have explored this interface and that's
been
just the tip of the iceberg of the
richness in terms of thinking
bi-directionally like in the
value-driven
motor control they didn't even mention
the environment they didn't mention
generative models of the niche
or having a second person who you're
playing with or an adversary or a
collaborator
it was very much internal and value
driven
and active inference blows the lid open
to play and collaboration and
a niche that's rich with rich models of
it
and the tight and reciprocal
relationship between
action and perception in active
inference resonates with other
key ideas related to embodied inactive
embedded and cultured extended eeee
approaches to cognition and agency any
adjective we like all the letters
but there's a whole area of sort of
critiques that

um swirled and eventually coalesced
that critiqued a sort of narrow
computationalist
framing of cognition and active
inference is almost
bringing these threads back together by
helping us have computational models
that respect the richness and the
embodiment in the world
but we're only steps down that
pathway in terms of where it can go
and in the end where the authors take us
is
towards a focus on interactions not
instructions and say that the
interactionist account portrays skill as
enacted without needing to assume it's
driven by instructions
that are implicitly couched in terms of
the content
of what is being represented you know
you can't say
pick up the arm so the knife is higher
without having a state estimate of where
the arm is and where the knife is
and their alternative interactionist
account for skilled performance
achieves this resolution
this improvement by avoiding two sets of
commitments
so for those who are commitment phobic
active inference is great for you
um it doesn't assume that generative
models used in active inference are the
models that are used by the organisms or
systems themselves
this is kind of the instrumental way of
thinking

we don't need to say that's what the organism is doing

we don't need to have a debate whether birds know where their wings are or what they think about what it's like to be a bird we can just have a framing that helps us understand how they act in the world and also the second commitment avoided by active inference

is and more importantly for their discussions about representation and instructions they do not assume that the explanatory story that's offered by sensory motor accounts needs to posit any causally efficacious mediating knowledge so you don't need to propose that there's some sort of kernel of knowledge that then gets enacted through

a command you have an account that just sidesteps that it doesn't say it doesn't exist it's just there's another way of framing what is happening so

i'll go to this closing slide and we can actually provide our last thoughts so what would a good understanding enable here we actually we didn't have the um

big questions and why it matters slide here because we kind of went through and through but um what would a good understanding enable like and this is an open question for those listening and also for the

authors like
why does it matter what would we be
enabling ourselves to do
if we had a better way or a different
way
of thinking about skilled performance
can everybody
play classical music or is there going
to be some other implication
what are the unique predictions and
implications so for those who
say hey i don't think omct is knocked
out yet
what is the differentiator what's the
experiment or the test
that we can actually do to see where
active inference
is coming into play what are the next
steps for us
as a research community and what are the
goals
of skilled performance research and of
course what are you still
curious about so any closing thoughts
here
well i'll just give one i want to thank
both of you guys for keeping this
grounded because
so many times i could have flown off on
this but one of the things i do want to
i'm going to want to ask the authors
about is i do think that there is a
place for
construction it's but it's a tiny
it's a fragment of the total memory
sphere and it's to do with perspective
memory
where if i don't remember to bring the

milk home

when i'm driving home from work i am

going to get in trouble

because i forgot to to carry out a side

action in the in the continuum of

i have to come home from work today but

i can't forget to bring

a quart of milk home with me so i do

think that there is a part for this

this instructionist piece in the memory

scape

but i don't think it should be the total

picture so i'm going to want to ask the

authors about that

perfect and that's a theme that we've

seen come up which is like

just because active inference is awesome

doesn't mean it's our only

tool in the toolbox so like even earlier

today we were hearing about how

active inference models can then be used

as input

into a just totally different kind of

model to

draw on the strengths of both so it

could be possible that

there's things that are described well

by active inference

and then there's a rapper that is almost

instructionist

so like people could give instructions

pick up your hand and then the person

does it

so was that not an instruction i mean

what does instruction mean if that's not

an instruction

so how do we combine active inference

models with other kinds of models blue

yeah i mean that's a great question too
and i i also think about
um i don't know really like where
where like the model fits in and
really just like what i was think what i
was talking about earlier about like
phenomenological
consciousness like if i intend to do
something is that
like do intentions matter do beliefs and
desires and intentions
like matter um sometimes but like also
like if
you know a tennis ball is flying at my
face and i reach my hand up and catch it
like without thinking
like that's like a very like i just
a perception action a perception action
like just happened all at the same time
right
so is there space for for the thought um
about the action
or is it always i don't know i'm still
squishy like
where does that we're just thinking
about doing something
fit into the active inference loop
because sometimes you think about it and
sometimes you don't
sometimes you you rehearse you know you
you uh think about think about it and
then you
choke or you don't other times there's
like i mean famously captured by
viral videos since the beginning of time
you know somebody does like an
incredibly
amazing maneuver that they certainly

didn't plan
and they didn't have knowledge of and
they probably could never do it twice
so it's awesome to have a framework that
covers potentially
our autonomic reflexes all the way
up perhaps to culturally infused and
trained performance
and enables us to think about how
pre-play and replay
of motor actions come into effect but
also those
america's home videos moments where
something
amazing just happens or something in
sports where something amazing happens
and you look at the replay it's like
what
how did that happen and the answer is
they didn't know it was going to happen
so right there it couldn't have been
knowledge driven
can i uh one last thing daniel for at
least for me yeah
again thanks blue for for putting in the
uh
the little clip about flow there because
i would like to find out if somebody
actually identifies by the sequence of
commands that they carried out like
if i carried out a set of instructions
would i identify
myself now as the carrier out of those
commands
or is my identity more about
those those aspects of
how i sort of move through
through time and change along with it

because if it is i guess if i could see
myself as being identified by
the instructions i carried out then i
guess optimal motor control theory
would be the whole thing but until
somebody shows me that that's
actually how i identify i think we re i
think we have to sort of
take it off that deflated sense and
inflate it up to the identity piece and
see the interstate for what it is
i'll close with a borjess quote inspired
by dean
time is the substance i am made of time
is a river which sweeps me along but i
am the river
it is a tiger which destroys me but i am
the tiger it is a fire which consumes me
but i am the fire
there you go perfect thank you everybody
for watching we hope to see you
participating soon